

# Ziyao (Richard) Cui

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## EDUCATION

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### Massachusetts Institute of Technology

*Ph.D. in Operations Research*

- Advisor: Rahul Mazumder

Cambridge, MA

*August 2026 – Present*

### Duke University

*B.S. in Computer Science & B.S. in Statistical Science (Minor in Economics)*

- GPA: 4.0/4.0 (Summa Cum Laude)
- Advisors: Jian Pei, Cynthia Rudin

Durham, NC

*August 2022 – May 2026*

### University of Oxford

*Study Abroad Program in Mathematics and Economics*

- Advisor: Seth Flaxman

Oxford, United Kingdom

*January 2025 – June 2025*

## PUBLICATIONS

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- **Ziyao Cui** and Jian Pei. How Is Uncertainty Propagated in Knowledge Distillation? *arXiv preprint arXiv:2601.18909*, 2026.
- **Ziyao Cui** and Edric Tam. Learning Graph Operators for Linear Node Regression.
- Sahoko Ishida, Mohammed Aheed Osman, **Ziyao Cui**, Uchenna Agu, Emily Becher, Gabriel Battcock, Daniel Hernandez, Duccio Piovani, Frances Knight, Seth Flaxman, and Kevin Tang. Mapping Subnational Vulnerability to Inadequate Micronutrient Intake using a Bayesian Small Area Estimation Framework. *arXiv preprint arXiv:2604.14971*, 2026. *Under review at the International Journal of Health Geographics*.
- **Ziyao Cui**, Minking Zhang, and Jian Pei. 2026. On Membership Inference Attacks in Knowledge Distillation. SIGKDD Explor. Newsl. 28, 1 (June 2026), 32–40. <https://doi.org/10.1145/3820356.3820359>
- **Ziyao Cui** and Edric Tam. Beyond the Laplacian: Interpolated Spectral Augmentation for Graph Neural Networks. *arXiv preprint arXiv:2511.11928*, 2025.
- **Ziyao Cui**, Minking Zhang, and Jian Pei. Learning to Attack: Uncovering Privacy Risks in Sequential Data Releases. *arXiv preprint arXiv:2510.24807*, 2025.
- **Ziyao Cui**, Erick Jiang, Nicholas Sortisio, Haiyan Wang, Eric Chen, and Cynthia Rudin. Revisiting Broken Windows Theory. *arXiv preprint arXiv:2509.16490*, 2025. *Under review at AAAI 2026*.

## HONORS AND AWARDS

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|                                                                                        |           |
|----------------------------------------------------------------------------------------|-----------|
| NSF Graduate Research Fellowship                                                       | 2026      |
| GEM Fellowship (Declined)                                                              | 2026      |
| Alex Vasilos Memorial Award (Excellence in Computer Science Research), Duke University | 2026      |
| Graduation with Highest Distinction in Computer Science, Duke University               | 2026      |
| Phi Beta Kappa Honor Society                                                           | 2025      |
| ASA Data Expo Challenge Professional Category, Honorable Mention                       | 2024      |
| Dean's List with Distinction, Duke University                                          | 2023-2026 |

## SERVICES

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**Reviewer:** *NeurIPS* (2025, 2026)

**Teaching Assistant:** Theory and Algorithms for Machine Learning (CS671), Fall 2024; Real Analysis I (MATH531), Spring 2025

INTERNSHIP EXPERIENCE

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|--------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| <b>Outlier AI</b><br><i>Coding Expert and Reviewer</i>                                                       | May 2024 – Present<br><i>Remote</i>     |
| <b>Shanghai Research Institute of Computing Technology Co.</b><br><i>Research Product Development Intern</i> | May 2024 – August 2024<br><i>Remote</i> |

SKILLS

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**Languages:** English, Mandarin, Spanish, German  
**Technical Skills:** Python, PyTorch,  $\text{\LaTeX}$ , Java, C, R, SQL, MATLAB  
**Professional Memberships:** American Statistical Association (ASA), Royal Statistical Association (RSS),  
Institute of Mathematical Statistics (IMS)